REPRESENTING MEANINGS WITH MATH

NLP Foundations & Semantic Modeling

Valerio Modugno | May 2026

PRIMARY OBJECTIVES

Appreciation

Understand how computational linguistics represents meaning in a machine-readable format.

Dual Tenets

Familiarize with Denotational Semantics and the Distributional Hypothesis (DH).

Formalization

Translate linguistic concepts into robust mathematical structures.

DEFINING "MEANING"

"What is meant by a word, text, concept, or action."

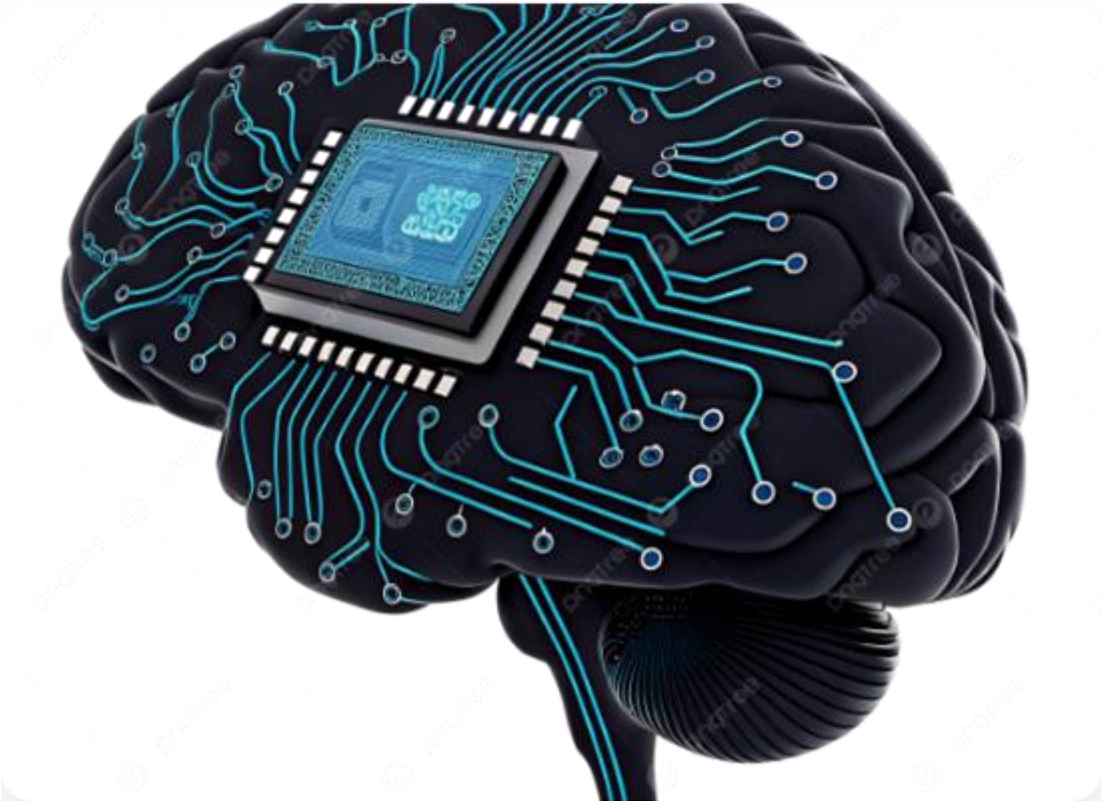
— Oxford English Dictionary

MEANINGS IN NATURAL LANGUAGE

Natural language is inherently meaningful, serving as the bridge between cognition and communication.

- Conveyance to an audience
- Target reference frames
- Symbols, social categories, and culture

MEANINGS IN NLP SYSTEMS



The Starting Point

Meanings are central to NLP analyses. How do machines 'look' at meanings? We must transition from human intuition to computational representation.

THE ANGLE OF COMPUTATIONAL LINGUISTICS

Modeling meanings in a mathematical framework

THE PERSPECTIVES PILLAR

Denotational Semantics

The relationship between signifiers (words) and what they stand for in reality (denotation).

Distributional Hypothesis

The assumption that semantic similarity correlates with distributional similarity in text.

DENOTATIONAL SEMANTICS INTUITION

Signifiers & Referents

Focuses on the salient features associated with an entity and the cognitive/behavioral effects of using a signifier.

Example: The word "Apple" denotes a specific fruit (object) or a technology company (corporate entity) based on its denotative anchor.

LEXEME: 'HIP-HOP'

Complex Denotation

This lexeme conveys meanings beyond a genre; it reflects values, norms, and beliefs orienting audience behavior.

DISTRIBUTIONAL HYPOTHESIS (DH)

"A difference of meaning correlates with a difference of distribution."

— Zellig Harris / J.R. Firth

DH serves as the theoretical framework for building computational models of **semantic memory**.

THE STANDPOINT OF MACHINES

Computability vs. Understanding

PERSPECTIVES CONFLICT

Feature	Human Beings	Machines
Language Nature	Entrenched in symbols & categories	Non-natural association
Sense Making	Innate & intuitive	Lacks inherent understanding
Computing	Bad at large datasets	Superior at scale & speed

HANDLING MEANINGS

Pattern-Matching

Humans provide rules to induce meaningful responses (e.g., RegEx).

Statistical Discovery

Frameworks allow machines to learn semantic similarity from data.

FORMALIZING DISCRETE SPACE

We define the Vocabulary V and represent word w_i as a vector in . $\mathbb{R}^{|V|}$

$$x = [0, \dots, 1, \dots 0]$$

The Orthogonality Problem

The dot product of *car* and *automobile* is exactly **0**. Mathematically, no two words share any similarity in traditional discrete space.

REGULAR EXPRESSIONS

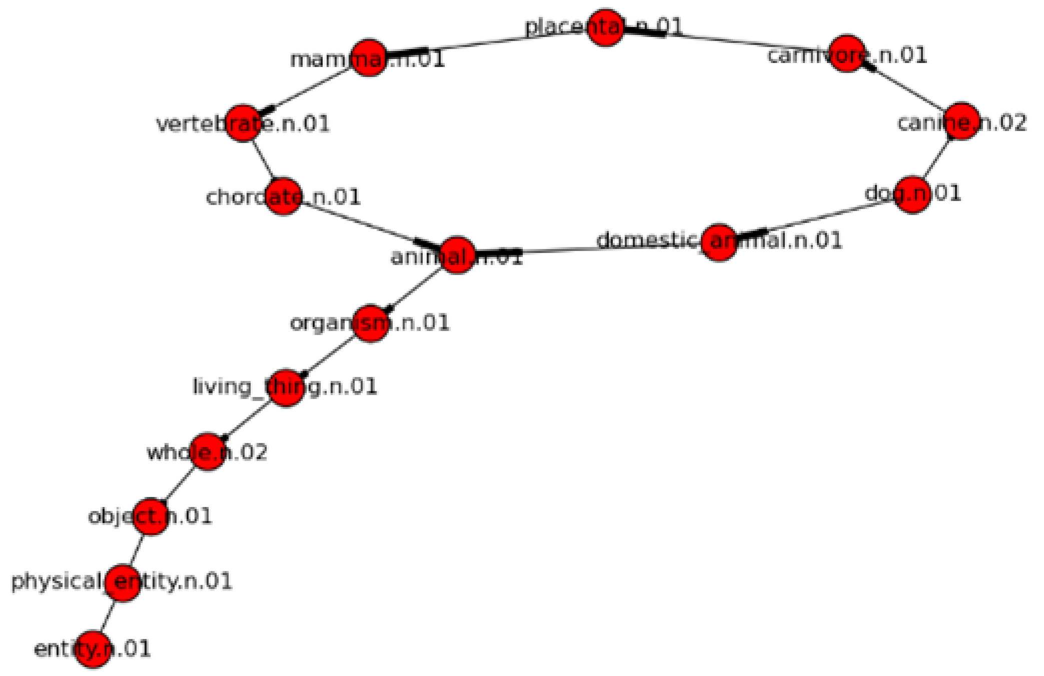
```
/buy|purchase\s+(item|product)/
```

Predictable behavior and flexible enough to power sophisticated engines like Amazon Alexa and Google Now.

DICTIONARY APPROACHES

WordNet®

A human-annotated lexical database grouping nouns, verbs, and adjectives into cognitive synonyms (**synsets**).



SUMMARY: PART 1

- Meanings are the core starting place for NLP.
- CL relies on Denotational and Distributional perspectives.
- Machines simulate understanding via learned data patterns.

MEANINGS IN HUMAN-ANNOTATED DICTIONARIES

Structured Semantics & WordNet®

PART 3: RULE-BASED PATTERN MATCHING

LEARNING OBJECTIVES

Familiarization

Understand the structure and cognitive groupings of WordNet, the premier human-annotated dictionary.

Scope of Application

Clarify how and when human-annotated dictionaries are deployed within broader NLP systems.

WHY DICTIONARIES MATTER

Bridging the Gap

Because machines cannot *naturally* associate meanings with lexemes, human intervention is required to explicitly map these connections.

Human-annotated dictionaries provide the necessary **'pattern-matching' rules** to induce meaningful, logical responses from machines.

Chatbots & Dialogue

Rules dictate safe, predictable logic flows in enterprise dialogue engines.

INTRODUCING WORDNET

More Than a Thesaurus

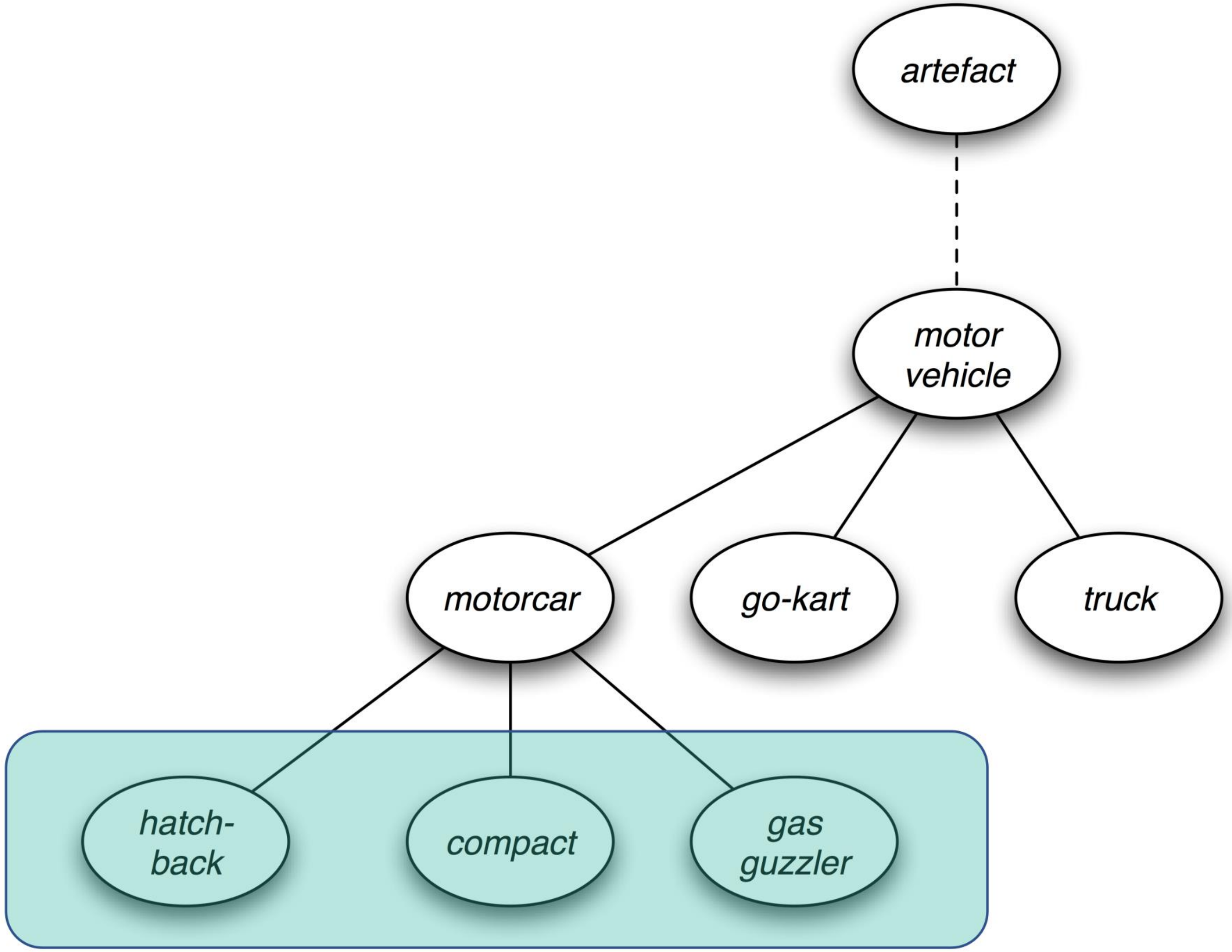
WordNet is the most popular human-annotated lexical database for the English language.

- Groups nouns, verbs, adjectives, and adverbs.
- Forms distinct cognitive clusters.
- Embeds critical semantic nuances beyond simple synonyms.

117k+

Distinct Synsets

A FRAGMENT OF THE WORDNET CONCEPT HIERARCHY



THE STRUCTURE OF WORDNET

The Synset

The primary relation is synonymy (e.g., 'shut' and 'close'). These are grouped into unordered, homogenous sets called **synsets**.

Conceptual Links

Synsets are not isolated; they are heavily interlinked by explicit conceptual relations, creating a massive semantic graph.

Glosses

Every synset contains a "gloss" (a dictionary-style definition) and often includes practical example sentences.

SUPER–SUBORDINATE RELATIONS

Hypernymy & Hyponymy

The most frequently encoded relationship in WordNet is the "ISA" (is-a) relation.

↑ **Hypernym:** General synset (e.g., Furniture).

↓ **Hyponym:** Specific synset (e.g., Bed, Bunkbed).

FURNITURE (Hypernym)

↕

BED (Hyponym)

↕

BUNKBED (Hyponym)

THE CONCEPT HIERARCHY

Mapping from Abstract Concepts down to Concrete Terms

Entity / Object

Artefact

Motor Vehicle

Car / Automobile

Hatch-back

NLP SYSTEMS & APPLICATIONS

Enhancing Chatbots

Allows conversational agents to recognize that a user asking for a "sofa" will also be satisfied with a "couch" or "settee".

Recommendation Systems

Enables e-commerce engines to suggest complementary items from related synsets (e.g., suggesting light bulbs when a lamp is purchased).

Moving Beyond Statistics

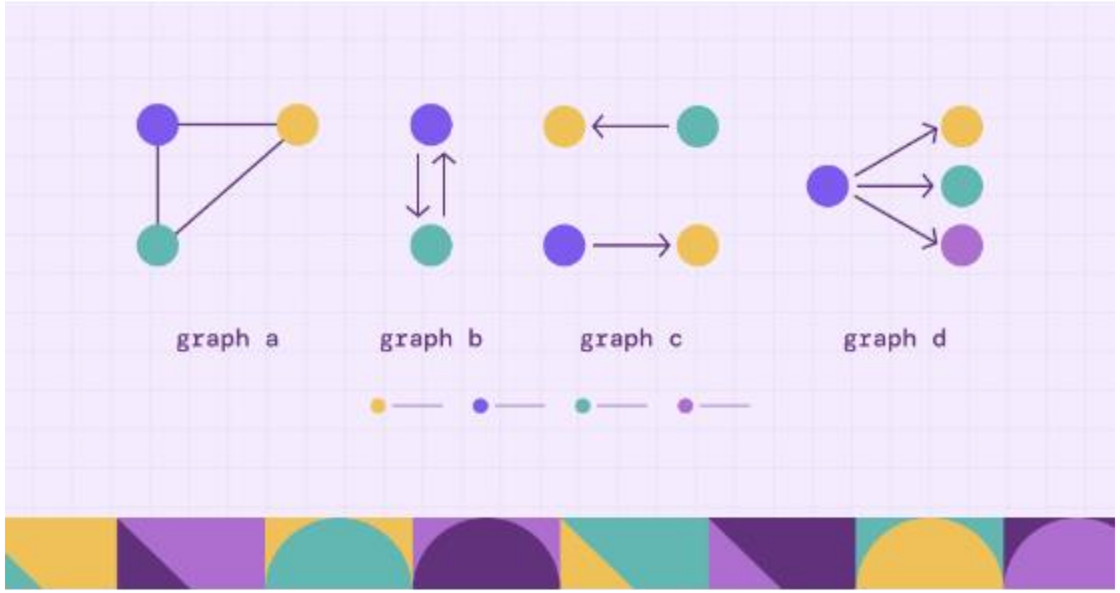
Vector Semantics (Stats)

Relies on massive corpora and co-occurrence patterns. It is "noisy," data-hungry, and often lacks a precise explanation for *why* two words are similar.

Graph Semantics (Logic)

Relies on human-curated knowledge (like WordNet). It is structured, logical, and explains similarity through established relationships like "is-a."

WordNet as a Knowledge Map



A Directed Acyclic Graph (DAG)

WordNet is not just a list; it is a **graph** where concepts are nodes and relations (like synonymy or hyponymy) are edges. Because these relationships flow in a specific direction (from general to specific), we can compute "paths" between any two concepts in the hierarchy.

Navigating the Tree

Depth

The distance from the root (the most general concept, like *Entity*) to a specific node.

Path

The shortest route between two nodes, passing through their shared parents.

LCS

The **Least Common Subsumer**—the first shared "ancestor" where two paths meet.

Quantifying Proximity

Wu-Palmer Similarity (WUP) calculates a score between 0 and 1 by looking at the depth of the shared ancestor relative to the depths of the two target words.

$$\mathrm{Sim}_{\mathrm{wup}}(s_1, s_2) = \frac{2 \cdot \mathrm{depth}(\mathrm{LCS})}{\mathrm{depth}(s_1) + \mathrm{depth}(s_2)}$$

Variables:

- **depth(s1):** Distance of word 1 from the root.
- **depth(s2):** Distance of word 2 from the root.
- **depth(LCS):** Distance of their shared ancestor from the root.

The closer the words are to their shared parent, the higher the score.

Worked Example: Car & Truck

0.88

Similarity Score

Calculation Breakdown:

Comparing **Car** and **Truck**:

- **Target 1 (Car):** Depth = 14
- **Target 2 (Truck):** Depth = 11
- **LCS (Motor Vehicle):** Depth = 11

Formula: $(2 \times 11) / (14 + 11)$ = **22 / 25 = 0.88**

Case Study: Apple & Orange

0.91

Similarity Score

Biological Consistency:

Even though an Apple and an Orange look different, WordNet knows they share a very specific "Parent":

🍎 **LCS:** Edible Fruit

Because "Edible Fruit" is deep in the hierarchy, the Wu-Palmer algorithm rewards their shared specific ancestry with a very high similarity score.

Summary: WUP vs. Vectors

Feature	Wu-Palmer (WUP)	Cosine Similarity (Vectors)
Primary Source	Human Dictionaries (WordNet)	Raw Text Frequencies
Mechanism	Graph Path & Depth	Angular Distance
Explainability	Very High ("They share a parent")	Moderate ("They share neighbors")
Handling Ambiguity	Explicit (Separate Synsets)	Implicit (Context Dependent)

Why use WUP?

Logical Reasoning

While vectors are excellent for "feel" and "style," Wu-Palmer is essential for **Knowledge Retrieval** and **Logical Disambiguation**.

It allows an AI to understand the rigid boundaries between concepts—ensuring it knows a *Compact Car* is definitely a *Motor Vehicle*, not just something that happens to be mentioned near roads.

IS WORDNET 'ENOUGH'?

The Value

Brings highly structured, highly usable meanings to machines. It remains a fantastic baseline resource for academic research and rule-based systems.

The Bottlenecks

- ✕ Requires massive, ongoing human labor.
- ✕ Difficult to keep up with evolving language.
- ✕ Subjectivity and missing nuance (e.g., treating 'proficient' and 'good' as perfect equals).
- ✕ Lacks a continuous, fluid measure of word similarity.

WRAP-UP: PART 2

Hierarchical Organization

WordNet effectively groups English words into homogenous synsets organized via deep hyponymy-hyperonymy relationships.

NLP Foundation

This structure is the backbone for traditional NLP systems that require strict, human-defined, rule-based meanings to function accurately.

MEANINGS IN STATISTICAL & ML MODELS

From Bag-of-Words to Learned Representations

PART 4: THE DATA-DRIVEN
SHIFT

LEARNING OBJECTIVES

Traditional vs. Modern

Appreciate how traditional NLP models utilize basic statistics while modern models leverage advanced Machine Learning.

Representing Meanings

Understand the shift from manually defined ontologies to representations learned directly from raw corpora.

THE FUNDAMENTAL QUESTION

"Is there a statistical or Machine Learning approach that might work in place of an annotated ontology or a pattern-matching method?"

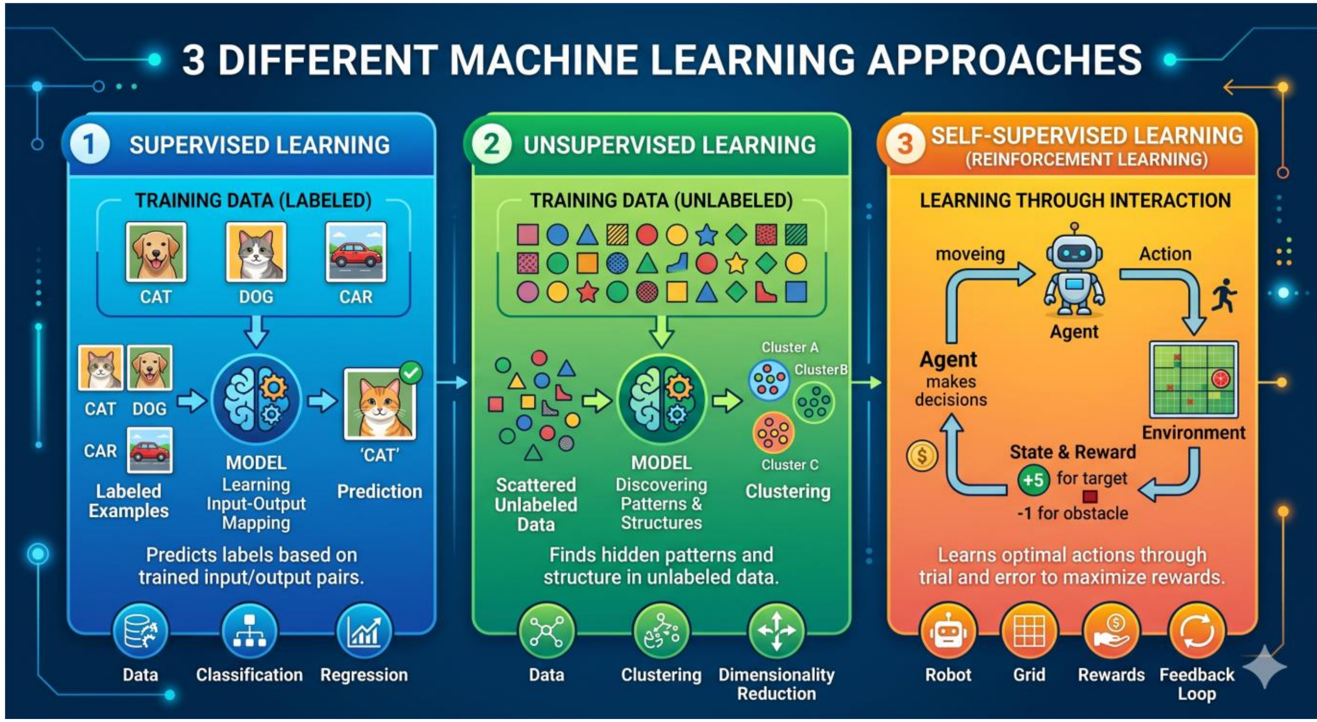
We explore the alternatives to human-intensive manual curation.

MACHINE LEARNING PARADIGMS

Foundational Logic

The choice of paradigm dictates how meanings are extracted and processed by modern architectures.

- **Supervised:** Learning from labels.
- **Unsupervised:** Finding hidden structures.
- **Self-Supervised:** Finding hidden structure by collecting labelled data



SUPERVISED LEARNING

Text Classification

Models like the Naive Bayes Classifier require a dataset where humans have provided labeled examples.

Sentiment Analysis

Example: Tagging a corpus of reviews as "Positive" or "Negative" to teach the machine semantic polarity.

UNSUPERVISED LEARNING

Topic Modeling

Discover thematic structures in massive, unlabeled corpora (e.g., financial filings).

Latent Dirichlet Allocation (LDA): Discovers hidden themes based purely on word co-occurrence patterns.

Hidden Themes

The machine groups words like {bank, interest, loan} into a single thematic cluster without human guidance.

TWO ROUTES FOR NLP

Traditional NLP

Heavily reliant on manual rules, dictionaries (WordNet), and simple frequency statistics.

Modern NLP

Driven by large-scale Machine Learning, neural networks, and dense vector embeddings.

TWO ROUTES FOR NLP

Traditional NLP

Heavily reliant on manual rules, dictionaries (WordNet), and simple frequency statistics.

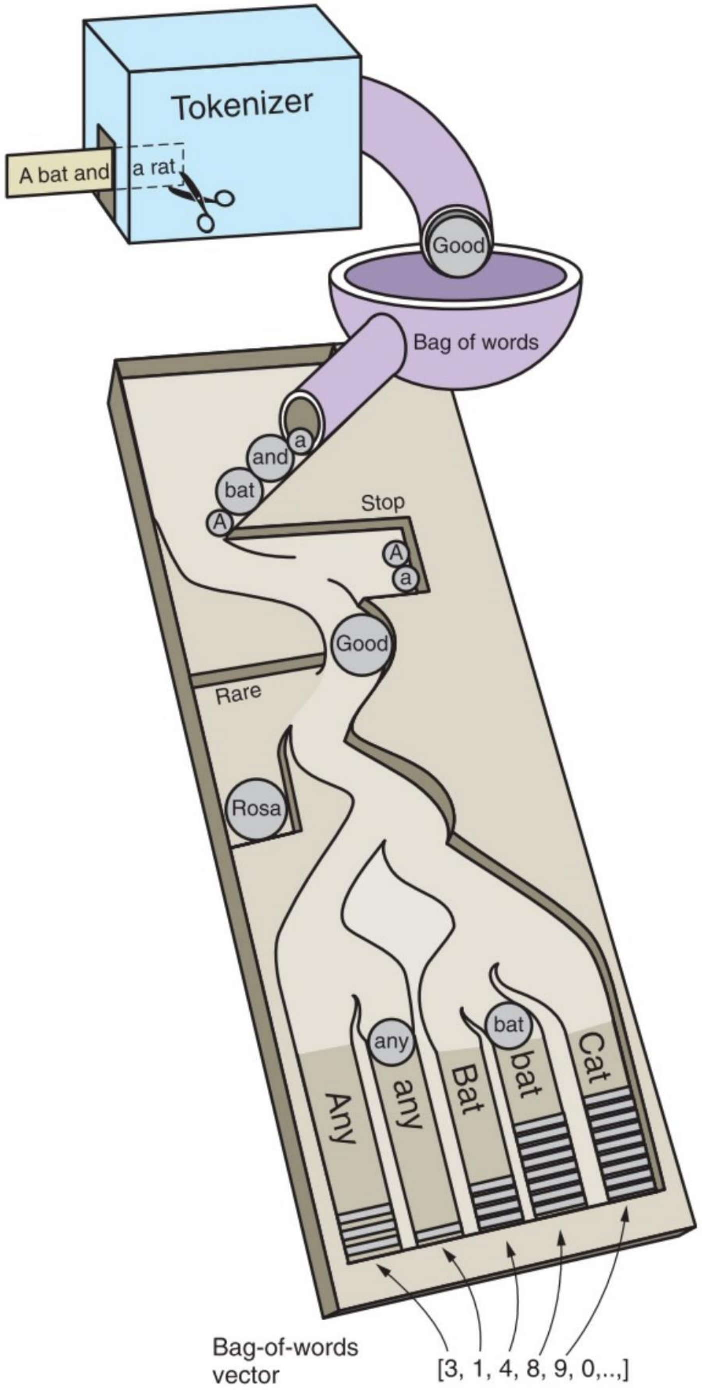
Mode

Driven e-scale M earni ural networks ense vector

THE BAG-OF-WORDS APPROACH

The workflow of Traditional NLP represents documents as a collection of their individual word frequencies.

- Tokenize documents in a corpus.
- Establish cardinality of unique items.
- Create a sparse vector space.



TOKENIZATION

Cutting Up the Text

A Tokenizer segments raw text strings into discrete units (tokens) to create the foundational BoW vector containing raw frequency counts.

This is the essential first step before any mathematical distance calculation can occur.

Algorithm 1 Bag-of-Words (BoW) Vectorization

Require: Corpus $\mathcal{C} = \{D_1, D_2, \ldots, D_M\}$
Ensure: Document-Term Matrix $\mathbf{X} \in \mathbb{N}^{M \times N}$

$\mathcal{V} \leftarrow$ Extract unique tokens from $\mathcal{C}$ {Define Vocabulary of size N}
Initialize $\mathbf{X}$ as a zero matrix of size $M \times N$
for each document $D_i \in \mathcal{C}$ **do**
    for each token $t \in D_i$ **do**
        if $t \in \mathcal{V}$ **then**
            $j \leftarrow$ index of t in $\mathcal{V}$
            $\mathbf{X}_{i,j} \leftarrow \mathbf{X}_{i,j} + 1$ {Increment Term Frequency}
        end if
    end for
end for
return $\mathbf{X}$

THE MATHEMATICAL BRIDGE

Vectorization

Bag-of-Words transforms raw text into a fixed-length numerical vector, mapping the linguistic world into a tensor space where classical math applies.

Feature Engineering

By treating each word index as a feature dimension, we enable algorithms to compute distances, weights, and decision boundaries across the lexicon.

Compatibility

This numeric representation serves as the universal interface for traditional ML libraries like Scikit-Learn, SciPy, and NumPy to process language.

Supervised Discovery

Learningthe Black Box relationship of a Labelled Corpus

MULTINOMIAL NAIVE BAYES

P(c|d)

The Probabilistic Baseline

The "Independence" Assumption

Naive Bayes assumes that every word in a "bag" is independent of every other word. This perfectly mirrors the BoW model's lack of syntax.

$$P(c|d) \propto P(c) \prod_{i=1}^{n} P(w_i|c)$$

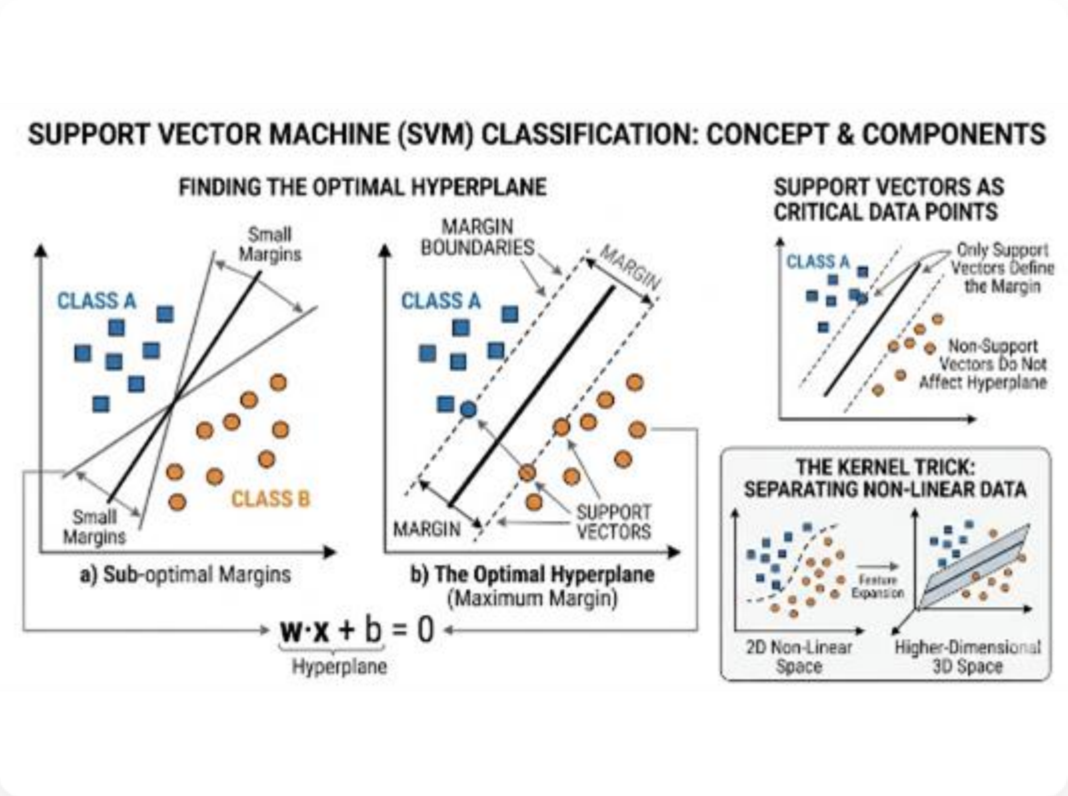
Highly effective for high-dimensional, sparse text classification tasks like Spam detection.

SUPPORT VECTOR MACHINES (SVM)

High-Dimensional Separation

SVMs are the gold standard for high-dimensional text classification. They find the **maximum-margin hyperplane** that separates word clusters.

- **Robustness:** Highly resistant to the "Curse of Dimensionality" in sparse BoW matrices.
- **Efficiency:** Linear SVMs scale effectively to vocabularies of 100k+ tokens.
- **Boundary:** Maximizes the "street width" between classes.



INTERPRETABLE LOGISTIC REGRESSION

Mapping Weight to Meaning

Logistic Regression learns a specific coefficient (β) for every word in the vocabulary. This makes the model uniquely interpretable for NLP.

$$z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$$

Feature Coefficients

By inspecting weights, we can see exactly which words trigger a classification:

- **Excellent:** High positive weight.
- **Worst:** High negative weight.

This allows for "Reasoning" behind the machine's decision-making process.

Unsupervised Discovery

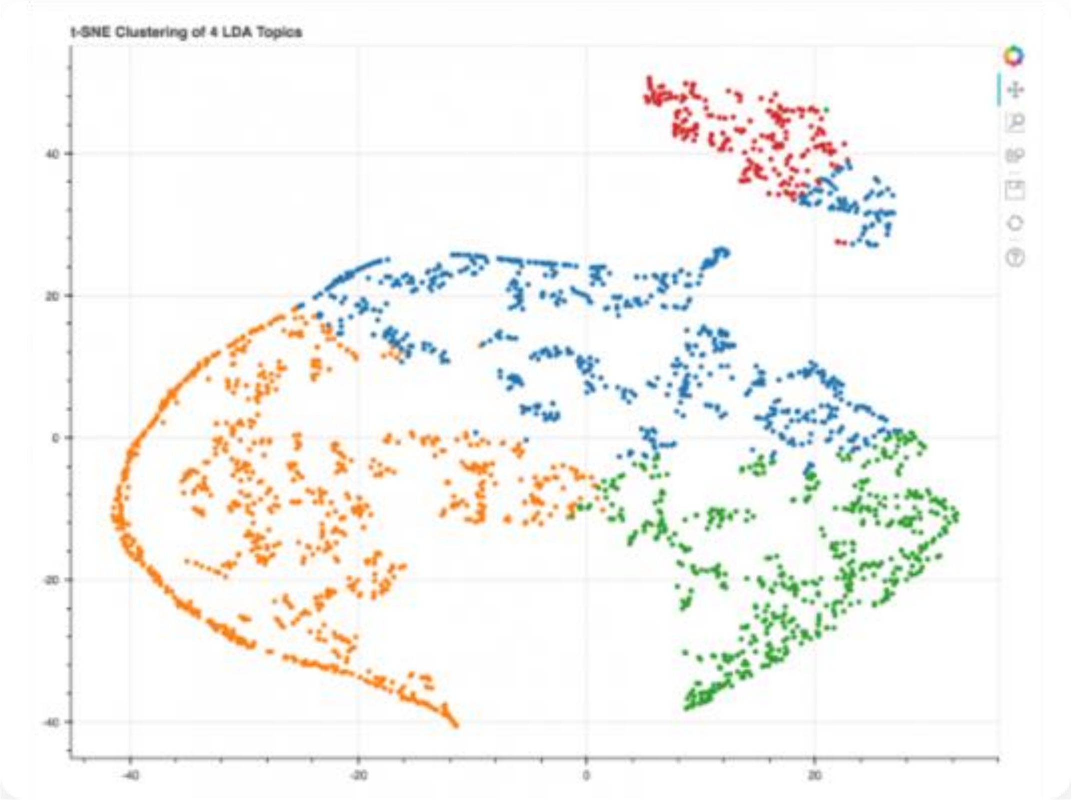
Automatically Revealing the Latent Structure of a Corpus

LATENT DIRICHLET ALLOCATION (LDA)

Topic Modeling Gold Standard

LDA assumes every document is a **mixture of topics**, and every topic is a **mixture of words**. It reverse-engineers the "Generative Process."

- **Global Statistics:** Uses co-occurrence to find word themes.
- **Clustering:** Groups words that frequently keep each other's company into latent themes.
- **Explainability:** Topics are represented as ranked word lists.

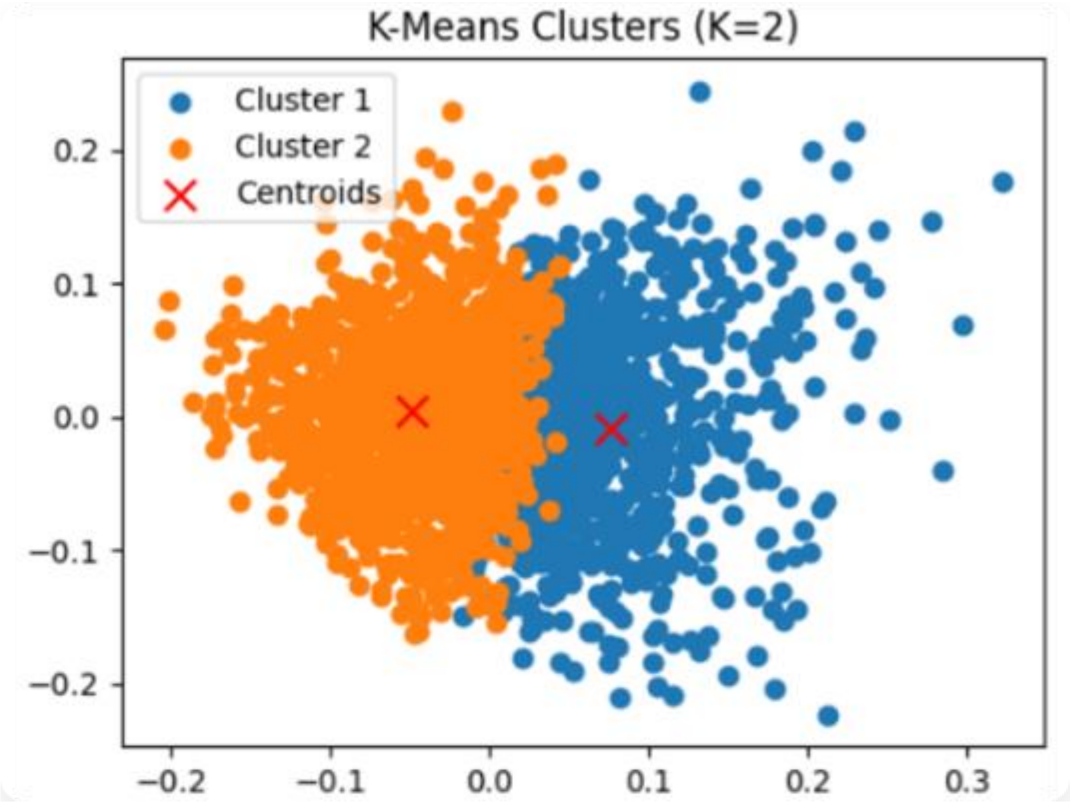


K-MEANS CLUSTERING ON BOW

Documents as Coordinates

Documents are plotted as points in high-dimensional vector space. K-Means finds the natural center (centroid) of these word clusters.

- **Geometric Logic:** Uses Euclidean or Cosine distance to measure similarity.
- **Macro View:** Organizes thousands of documents into distinct table-of-contents buckets.
- **Use Case:** Search engine grouping or news aggregation.



CHOOSING THE RIGHT ALGORITHM

Algorithm	Type	Strengths	Ideal Use Case
Naive Bayes	Supervised	Speed, Sparse Data Support	Spam Detection
SVM	Supervised	High-Dim Resilience	Complex Classification
LDA	Unsupervised	Theme Discovery	Topic Modeling
Logistic Reg.	Supervised	Interpretability	Sentiment Analysis

THE "MAN BITES DOG" FALLACY

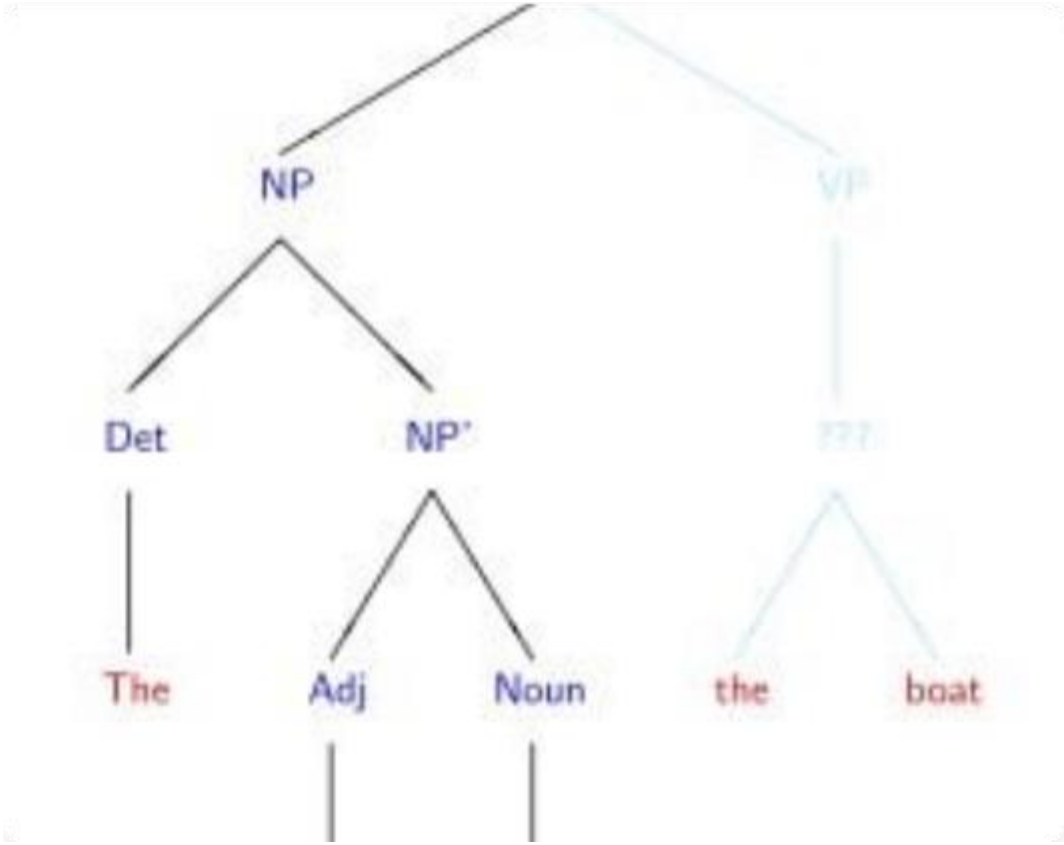
Loss of Compositionality

The primary weakness of BoW is the destruction of syntax. Because it relies on the **commutative property of addition**, word order is mathematically irrelevant.

Identical Vectors:

$${}_1(\text{Man bites dog}) = {}_2(\text{Dog bites man})$$

Meaning is emergent from sequence; by reducing a sentence to a "bag," we lose the relationship between agent and object.



THE EXPONENTIAL WALL

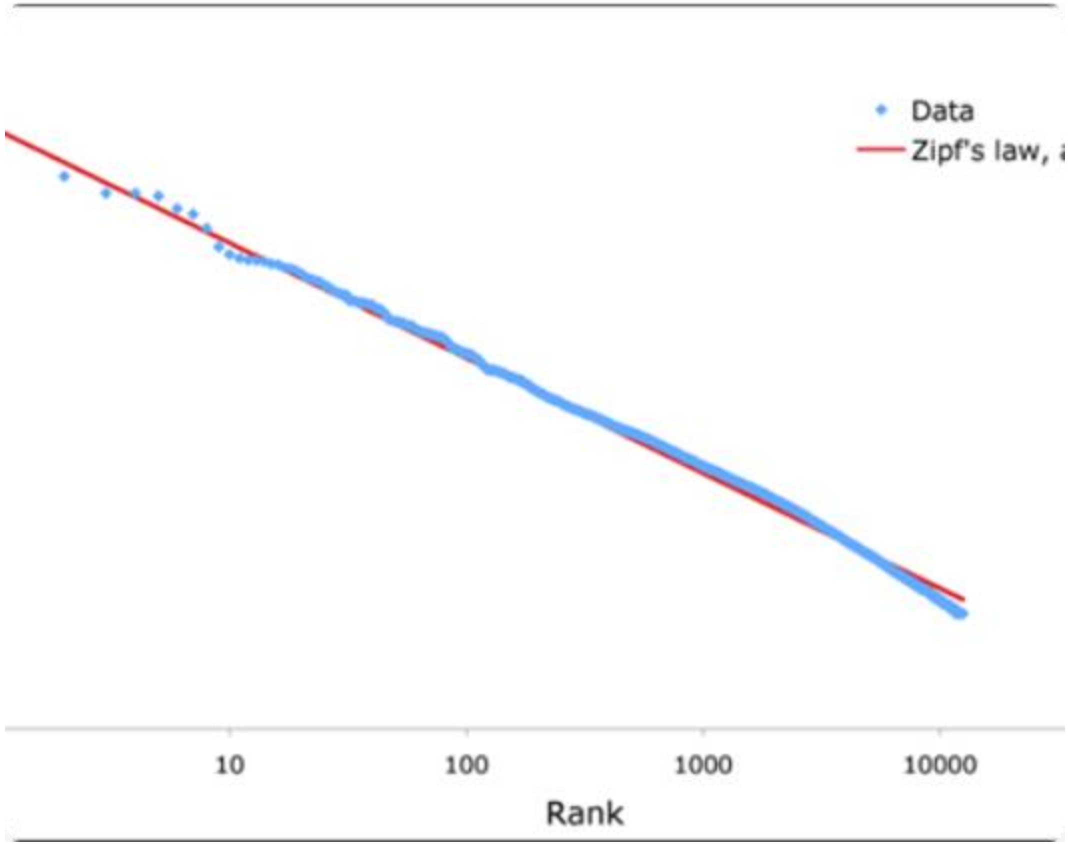
$|V|^2$

Dimensionality Explosion

The N-Gram Paradox

To fix the "Order Problem," researchers often use N-grams (pairs or triplets of words). However, this triggers a combinatorial explosion. If $|V| = 50,000$, a **Bigram** model expands to **2.5 Billion** dimensions. This makes the matrix too sparse to compute and too massive for modern RAM.

ZIPF'S LAW AND THE NOISE FLOOR



Statistical Imbalance

Natural language follows a **Power-Law Distribution**. The most frequent tokens (stop-words) carry the least semantic weight.

- **Sparsity:** Semantic-rich words appear too rarely to build robust statistical counts.
- **Dominance:** High-frequency "noise" (e.g., the, is, at) dominates the vector's magnitude.

This creates a critical need for weighting schemes like TF-IDF just to find the "signal" within the counts.

THE CURSE OF DIMENSIONALITY

50k+

Vector Dimensions

Proliferation of Unique Vectors

As unique lexical items grow, document vectors become massive and extremely sparse. In a raw corpus, most entries in the matrix are **zeros**.

One-hot vectors mitigate this by noting only presence (1) or absence (0), but information remains highly distributed across a space that is too large to compute efficiently.

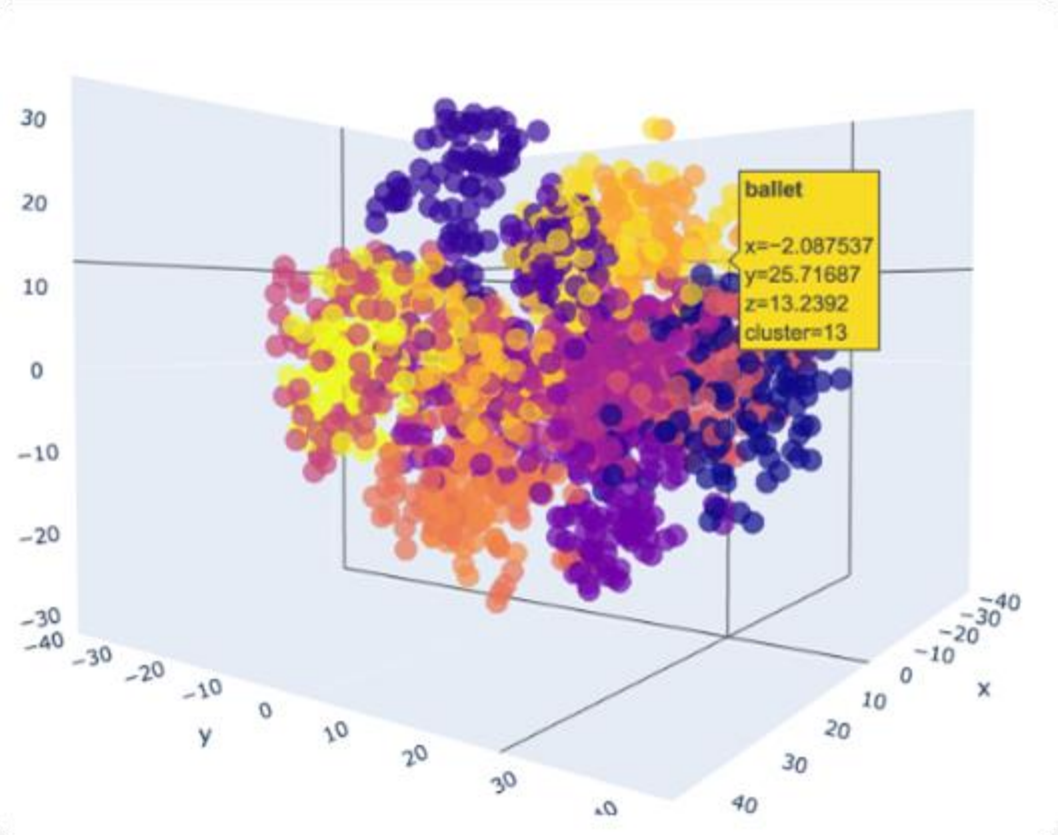
SEMANTIC FAILURE OF BOW

The Orthogonality Trap

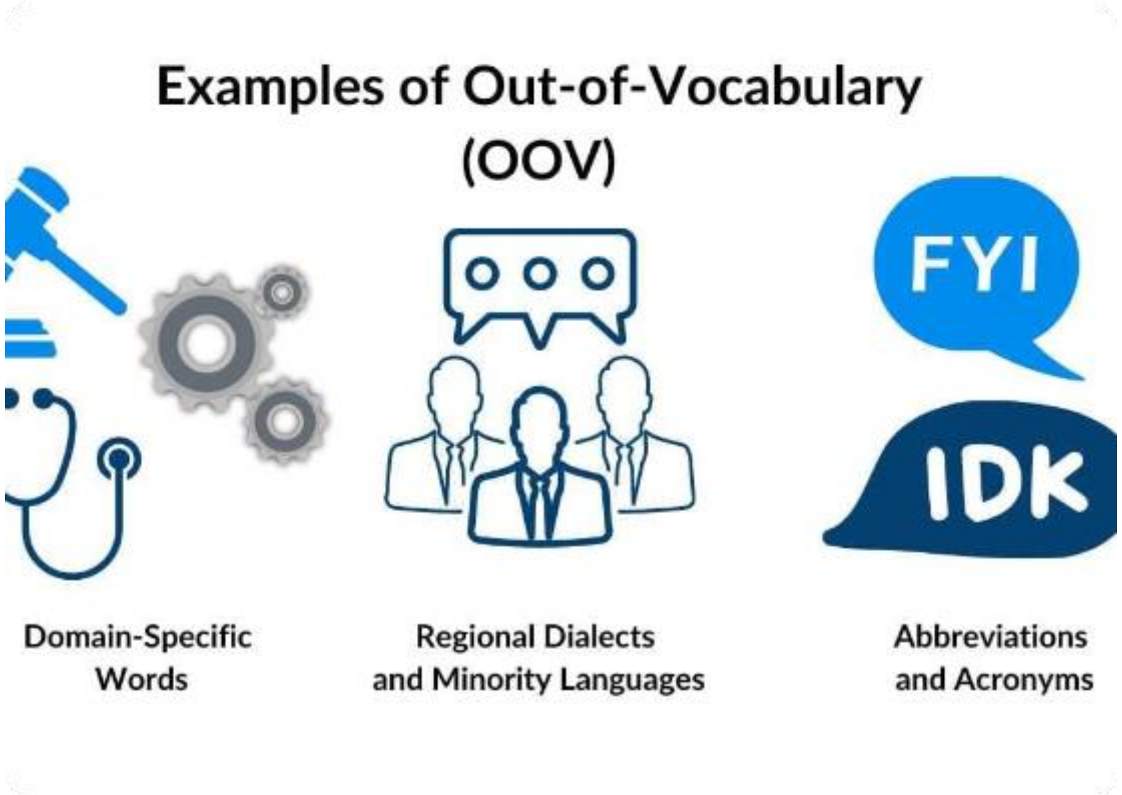
The most profound limitation is the **lack of semantic overlap**. One-hot vectors fail to capture relationships between synonyms.

- **'Good' vs 'Fine'** = Orthogonal (0 similarity).
- **'Greasy Spoon' vs 'British Cafe'** = Empty Intersection.

Because every unique word is its own dimension, the machine cannot perceive that two different words might point to the same concept.



THE CLOSED-WORLD PROBLEM



Out-of-Vocabulary (OOV)

BoW requires a fixed vocabulary defined during the training phase. It operates under a "Closed-World" assumption.

When a novel token appears in the real world (slang, new brands, acronyms), the model is literally **blind** to it. The matrix has no index for the new word, forcing it into a generic bucket and losing all signal.

BOW VS. THE REQUIREMENTS OF MEANING

Core Requirement	Bag-of-Words (BoW) Status	Modern Solution
Syntax & Order	❌ Destroyed (Commutative)	RNNs / Transformers
Polysemy	❌ Single Vector Per Word	Contextual Embeddings
Synonymy	❌ Orthogonal Vectors	Latent Space / Dense Vectors
Efficiency	❌ High-Dim & Sparse	Low-Dim & Dense

WRAP-UP: PART 3

- BoW is the quintessential traditional NLP model; it is intuitive but computationally demanding.
- Supervised learning provides specific classification, while Unsupervised learning discovers latent themes.
- **Critical Flaw:** BoW models miss the semantic nuance of synonyms and context, setting the stage for Part 5.

MEANINGS AND DISTRIBUTIONAL REPS

Vector Space Semantics & Modern Computational Modeling

PART 2: FROM HYPOTHESIS TO MATH

LEARNING OBJECTIVES

Consolidation

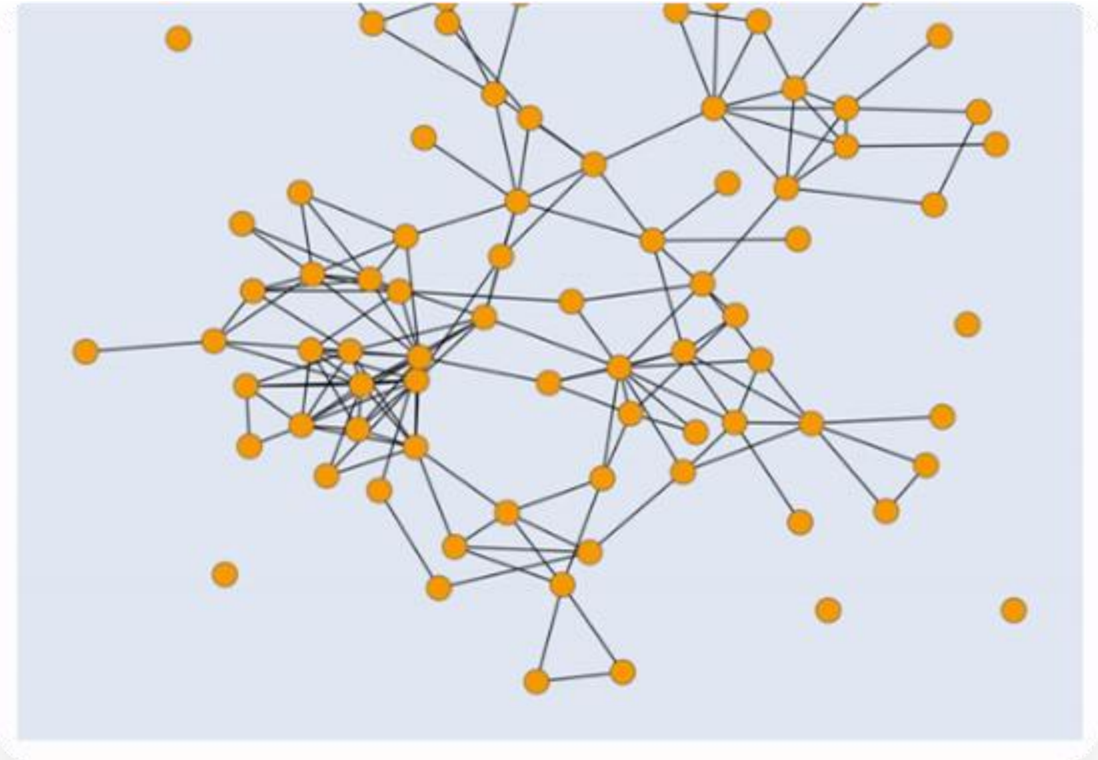
Solidify the 'Distributional Hypothesis' as the bridge between raw text and semantic meaning.

Inference

Apply mathematical operations to vectors to infer semantic similarity between disparate words.

THE DH RECAP

> *"Semantic similarity is a function of the contexts in which words are used."*
>
> — J.R. Firth (1957)

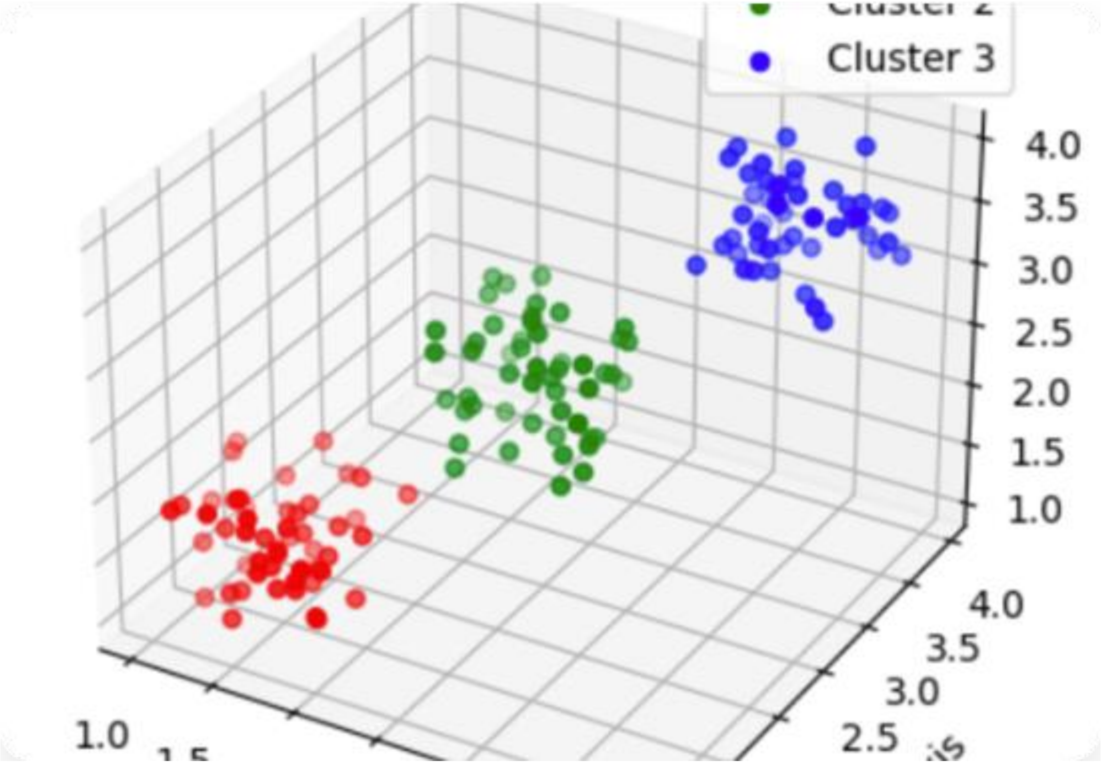


VECTOR SPACE SEMANTICS

Encoding Context

Each lexical item is mapped to a **distributional vector** representing its co-occurrence frequencies with linguistic contexts.

- High-dimensional spaces
- Latent semantic properties



LINGUISTIC CONTEXT TYPES

Context Type	Mechanism	Extracted Feature
Undirected window-based	Symmetric words in a N-word radius	Surface word collocates
Directed window-based	Separates Left vs. Right context	Positional collocates
Dependency-typed	Parses grammatical structure	Subject/Object dependencies

THE 3-STEP PROCEDURE

1. Extraction

Extract target-context co-occurrences from the corpus and count them.

2. Transformation

Convert raw counts into significance weights (e.g., PPMI, TF-IDF).

3. Measurement

Compute semantic similarity using vector comparison (e.g., Cosine).

STEP 1: CO-OCCURRENCE MATRIX

Scenario: Targets (bike, car, dog, lion) vs. Contexts

TARGET	bite	buy	drive	eat	ride
bike	0	15	0	0	85
car	0	42	91	0	5
dog	35	2	0	12	0
lion	12	0	0	45	0

*Fictional raw counts for illustrative purposes.

INFORMATION THEORY & TF-IDF

Scaling Entropy

Raw counts are biased toward common words. TF-IDF penalizes words that appear everywhere, highlighting unique significance.

$$IDF(t, D) = \log\left(\frac{|D|}{|\{d \in D : t \in d\}|}\right)$$

Intuition: Use a logarithmic scale to dampen the impact of frequent terms, forcing the machine to care about rare, informative lexemes.

STEP 2.0: WEIGHTING WITH PPMI

Positive Pointwise Mutual Information PPMI measures if target-context pairs co-occur more than expected by chance.

$$\mathrm{PPMI}(t, c) = \max\left(0, \log_2 \frac{p(t, c)}{p(t)\,p(c)}\right)$$

Zeros are kept for independent pairs; positive values indicate a semantic link.

STEP 2.1: WEIGHTING WITH PPMI

TARGET	bite	buy	drive	eat	get	live	park	ride	tell
bike	0	0.50	0	0	0	0	1.09	1.79	0
car	0	0.80	1.56	0	0	0	0.18	0	0
dog	0	0	0	2.01	0	1.65	0	0	2.16
lion	2.75	0	0	0	0.26	1.01	0	0	0

Pairwise Distributional Similarity

The Computational Comparison

Up to this point in the NLP pipeline, we have successfully converted linguistic items into **distributional vectors** based on their co-occurrences in a corpus.

Pairwise similarity is the process where the algorithm takes a single target word's vector and mathematically compares it against the vector of every other word in the vocabulary.

This ensures that we evaluate the semantic relationship across the entire lexical field, leaving no word unexamined.

Cosine Similarity Formula

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \|v\|}$$

For a vocabulary of 10,000 words, the system calculates 10,000 distinct similarity scores for each target.

THE GEOMETRY: COSINE VS EUCLIDEAN

$$\cos(u, v) = \frac{u \cdot v}{\| u \| \; \| v \|}$$

Why not Euclidean?

Euclidean distance (L_2) is sensitive to document length. A 100-page book and its 1-page summary would be distant due to magnitude alone.

The Cosine Solution

Cosine normalizes by length, evaluating only the **angle**. It focuses on relative proportions of words, i.e., pure semantics.

STEP 3: ASSESSING THE DISTRIBUTIONAL SIMILARITIES BETWEEN LEXEMES

The distributional similarity between two lexemes u and v is measured with the similarity between their distributional vectors u and v.

Once we have computed the pairwise distributional similarity between the targets, we can identify the k nearest neighbours of each target t, the k lexical items with the highest similarity score with t.

	TARGET			
	bike	car	dog	lion
bike	1			
car	0.16	1		
dog	0	0	1	
lion	0	0	0.17	1

MECHANICS OF K-NEAREST NEIGHBORS

The 'k' Parameter

In machine learning, **k** acts as a placeholder or threshold chosen by the researcher (e.g., 3, 5, or 10) to define the scope of the neighborhood.

Scoring & Sorting

Once the machine computes scores against the entire dictionary, it performs a **descending sort** from the highest similarity to the lowest.

Neighbor Identification

Identifying the **k-nearest neighbors** simply means selecting the top 'k' items from the sorted results—the semantic elite of the vocabulary.

CASE STUDY: TARGET "CAR" WITH K=3

Lexical Item	Similarity Score	Rank
automobile	0.92	1
van	0.88	2
truck	0.85	3
...	...	...
apple	0.01	9,842

The Algorithm's Choice

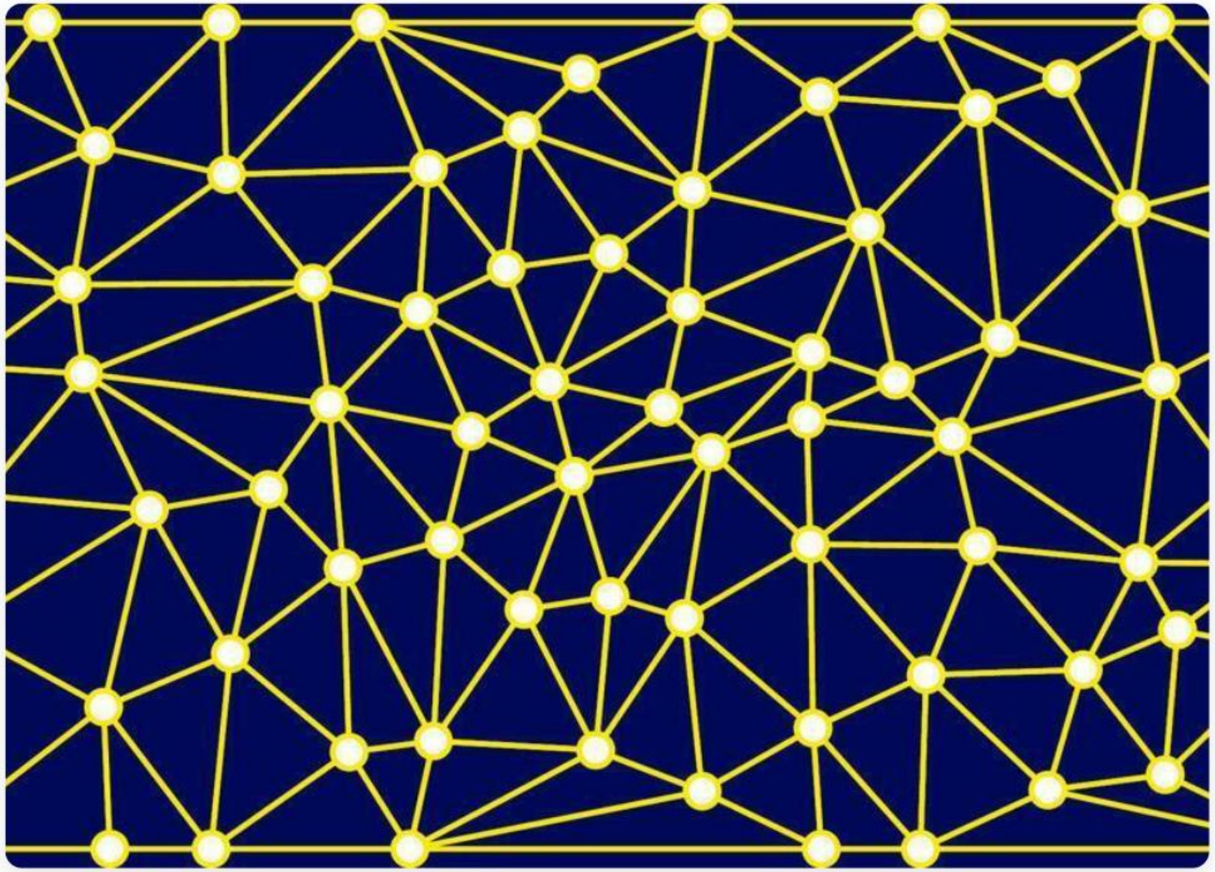
In this scenario, the algorithm identifies the top three items as the most distributionally similar to our target.

The machine effectively ignores "apple" due to its near-zero similarity, focusing only on the **semantic cluster** that shares functional contexts with "car".

Result Set (k=3):

{ automobile, van, truck }

THE COMPUTATIONAL PAYOFF



Data-Driven Intelligence

This is the ultimate payoff of the **Distributional Hypothesis**. By automating the identification of neighbors, the machine builds a **dynamic, self-learning thesaurus**.

Unlike human-annotated dictionaries (e.g., WordNet), this method requires no manual labor to define that a "car" is like a "van".

The machine **infers** this relationship because both words share identical distributional contexts: they both co-occur with verbs like *drive*, *park*, and *buy*.

Micro vs. Macro Views

Micro: k-Nearest Neighbors

Focuses on an **automated thesaurus** for a single word. It looks at the local neighborhood of a target vector to find immediate synonyms (e.g., car → automobile).

Macro: K-Means Clustering

Steps back to observe the **entire vocabulary** at once. It groups all vectors into broad thematic regions, revealing the latent organization of the whole lexicon.

Segmenting the Lexicon

10k

Word Dictionary

The "Macro" Question

If we have a massive 10,000-word dictionary represented in high-dimensional vector space, how can we organize it effectively?

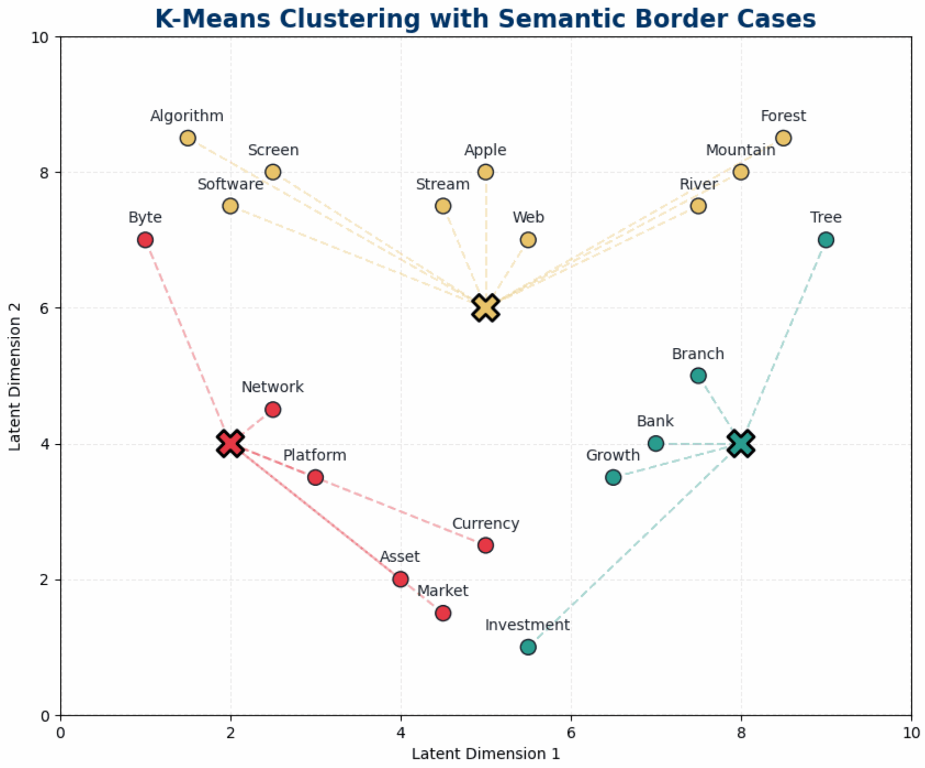
The goal is to divide this entire dictionary into, for example, **50 broad categories** based on natural boundaries in the data rather than human intuition.

The Emergence of Natural Boundaries

Automated Grouping

The algorithm groups the entire space into distinct semantic clusters without manual supervision.

- **Vehicles:** Car, truck, and bike are pulled into a single cluster.
- **Animals:** Lion, dog, and cat form a distinct biological group.
- **Finance:** Bank, money, and asset align in a business-centric cluster.



Use Case: Topic Discovery

Massive Datasets

Automatically identifies themes in millions of documents where manual labeling is impossible.

Social Media

Scraping and clustering tweets to see real-time shifts in public conversation and trending topics.

Theme Extraction

Discovers the "latent" thematic structure of a corpus based purely on word co-occurrences.

Resolving Ambiguity

Word Sense Disambiguation (WSD)

The Challenge

Tokens like **"bank"** are ambiguous. How can a machine know if we mean a financial institution or the side of a river?

The Clustering Solution

If we cluster the *context vectors* of every occurrence of "bank," the algorithm naturally separates them into two distinct clusters.

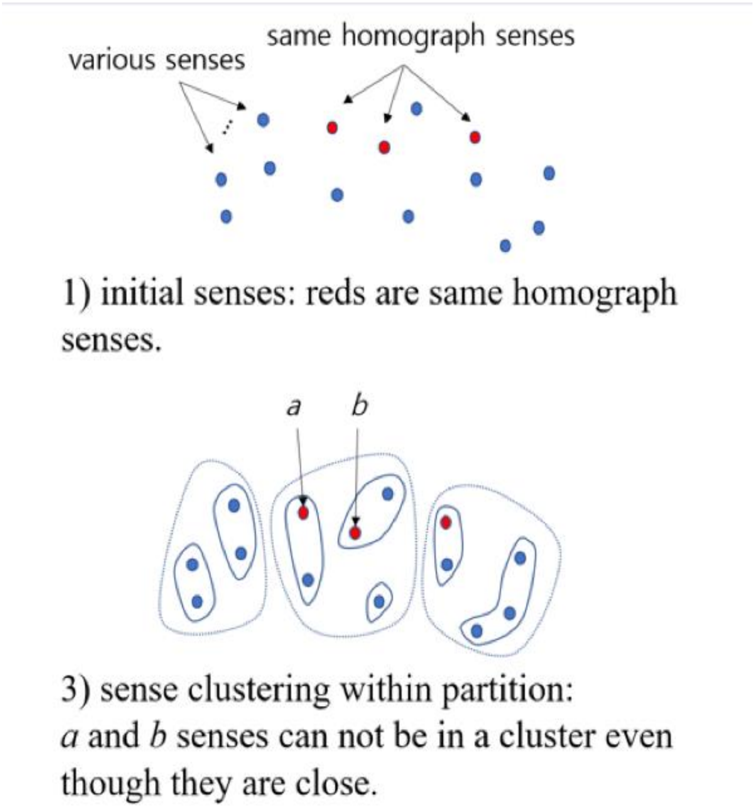
Visualizing the Sense Split

Contextual Separation

When K-Means analyzes the contexts, two "senses" emerge as separate clusters:

- **Sense A:** Associated with *money*, *vault*, *deposit*, and *account*.
- **Sense B:** Associated with *water*, *river*, *stream*, and *fishing*.

One word, two distinct mathematical locations.



Summary: Thesaurus vs. Table of Contents

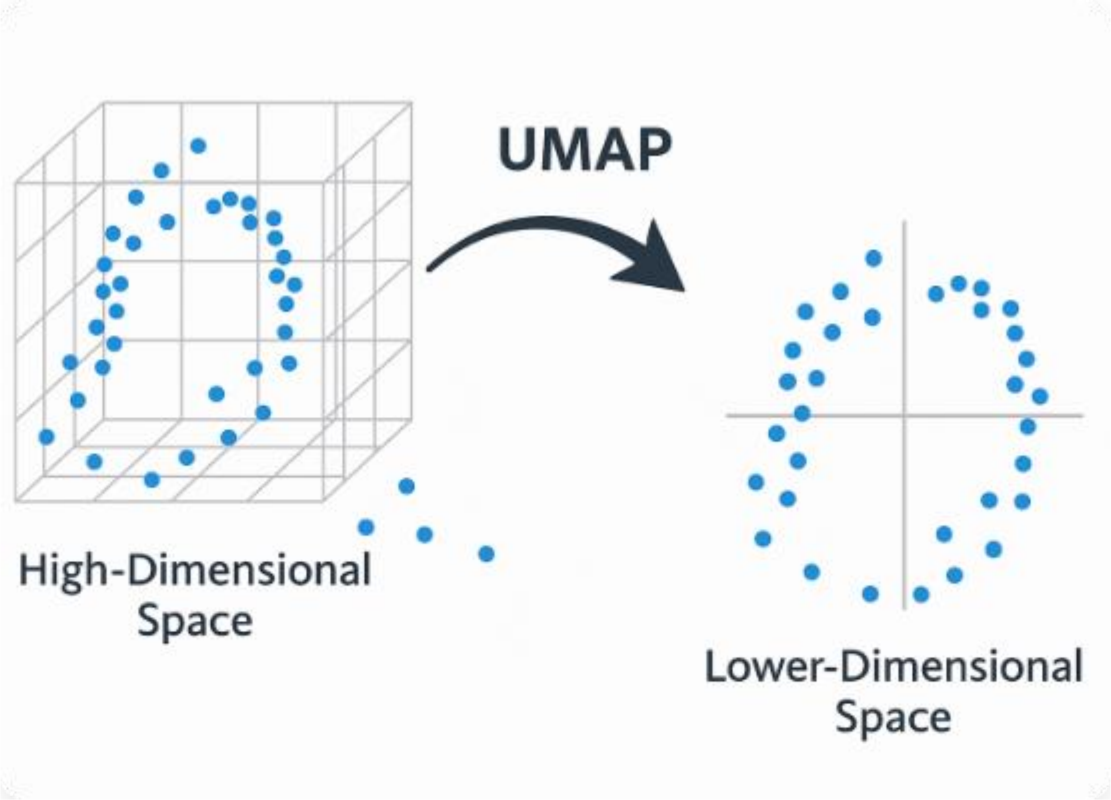
Feature	k-Nearest Neighbors	K-Means Clustering
Perspective	Micro (Local)	Macro (Global)
Metaphor	Automated Thesaurus	Automated Table of Contents
Input	One target word	The entire vocabulary/corpus
Primary Use	Search & Synonym retrieval	Topic Discovery & Ontology building

The Curse of High Dimensionality

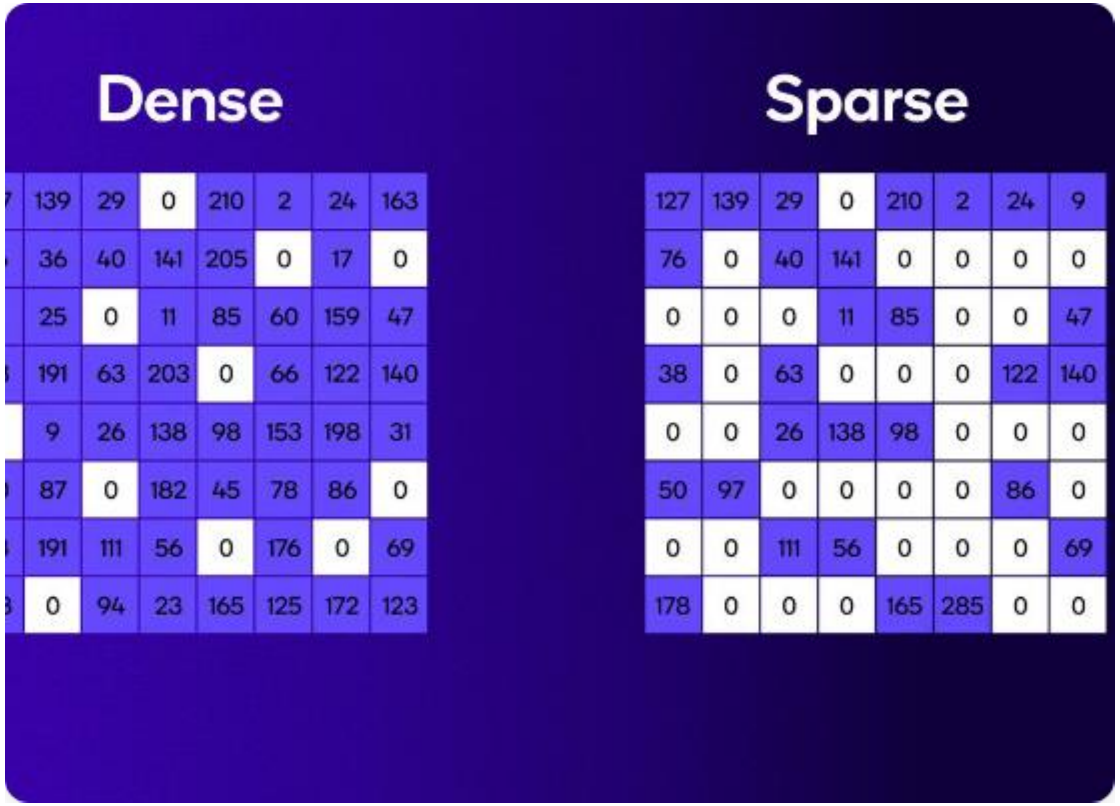
Massive Vector Spaces

When we represent words as vectors of their neighbors, the dimensionality of the space is equal to the size of our vocabulary ($|V|$). In a typical corpus, this can exceed 100,000 dimensions.

- **Computational Inefficiency:** Comparing vectors in 100k+ dimensions is extremely slow.
- **Data Sparsity:** Most entries in a raw co-occurrence matrix are **zeros**.
- **Storage:** Storing these matrices requires massive memory footprints.



From Sparse to Dense Representations



Compressing Information

Dimensionality reduction aims to transform high-dimensional, **sparse** vectors (mostly zeros) into low-dimensional, **dense** vectors (continuous real numbers).

By moving from 100,000 dimensions to just 300, we "force" the machine to find the most critical semantic features that distinguish words.

This process is not just about compression; it is about **abstraction**.

Latent Semantic Analysis (LSA)

Revealing the hidden conceptual structure of language

The Core Intuition of LSA

Words as Concepts

LSA assumes that words that appear in similar documents are likely related in meaning. It treats a document not just as a bag of words, but as a "bag of concepts."

Latent Meaning

By analyzing patterns of co-occurrence, LSA identifies "latent" (hidden) semantic structures. It can recognize that "Doctor" and "Physician" are related even if they never appear in the same sentence.

SVD: The "Three-Way Split"

Mathematically, SVD decomposes the original co-occurrence matrix M into three distinct components:

$$M = U \Sigma V^T$$

U Matrix

Relates **Words** to latent concepts. Each row is a word vector in "concept space."

$ \Sigma$ Matrix

Contains **Singular Values.** Represents the "strength" or importance of each latent concept.

V^T Matrix

Relates latent **Concepts** back to the original contexts (or documents).

Understanding the SVD Decomposition

Matrix	Formal Name	Linguistic Interpretation
U	Left Singular Vectors	The "semantic fingerprints" of words.
Σ	Singular Values	Weighting of how much each concept explains the data.
V^T	Right Singular Vectors	Mapping of how concepts are distributed across documents.
U_k	Truncated Word Space	The final dense embeddings used for semantic tasks.

Truncation: Finding Signal in Noise

k

Top Dimensions

Why Throw Away Data?

The "magic" of SVD happens when we keep only the top k dimensions (e.g., $k=300$) and discard the rest. The smaller singular values in Σ usually correspond to **noise** or idiosyncratic word choices in the corpus.

Truncation forces words that share partial contexts to be mapped into the exact same dimensions, effectively **merging synonyms** in vector space.

How LSA Solves Synonymy

Scenario:

"Doctor" and "Physician" rarely appear in the same document. In a raw matrix, their similarity is **Zero** (Orthogonal).

Result with LSA:

Because both "Doctor" and "Physician" co-occur with words like *hospital*, *patient*, and *surgery*, LSA places them in the same latent "Medical" dimension.

- ✓ Captures higher-order co-occurrence.
- ✓ Reduces sensitivity to exact word choice.
- ✓ Provides a compact, computable representation.

A Comparative Look at Representations

LSA, HAL, and GloVe: Different paths to semantic truth

LSA vs. HAL: Matrix Architectures

LSA (Latent Semantic Analysis)

Uses a **Term-Document** matrix. It captures "Global" context—the broad themes of a document. It's excellent for topic discovery but can miss local syntactic relationships.

HAL (Hyperspace Analogue)

Uses a **Term-Term** matrix. It focuses on "Local" context windows (e.g., 5 words). It excels at capturing grammatical and immediate semantic neighbors but is easily biased by very frequent words.

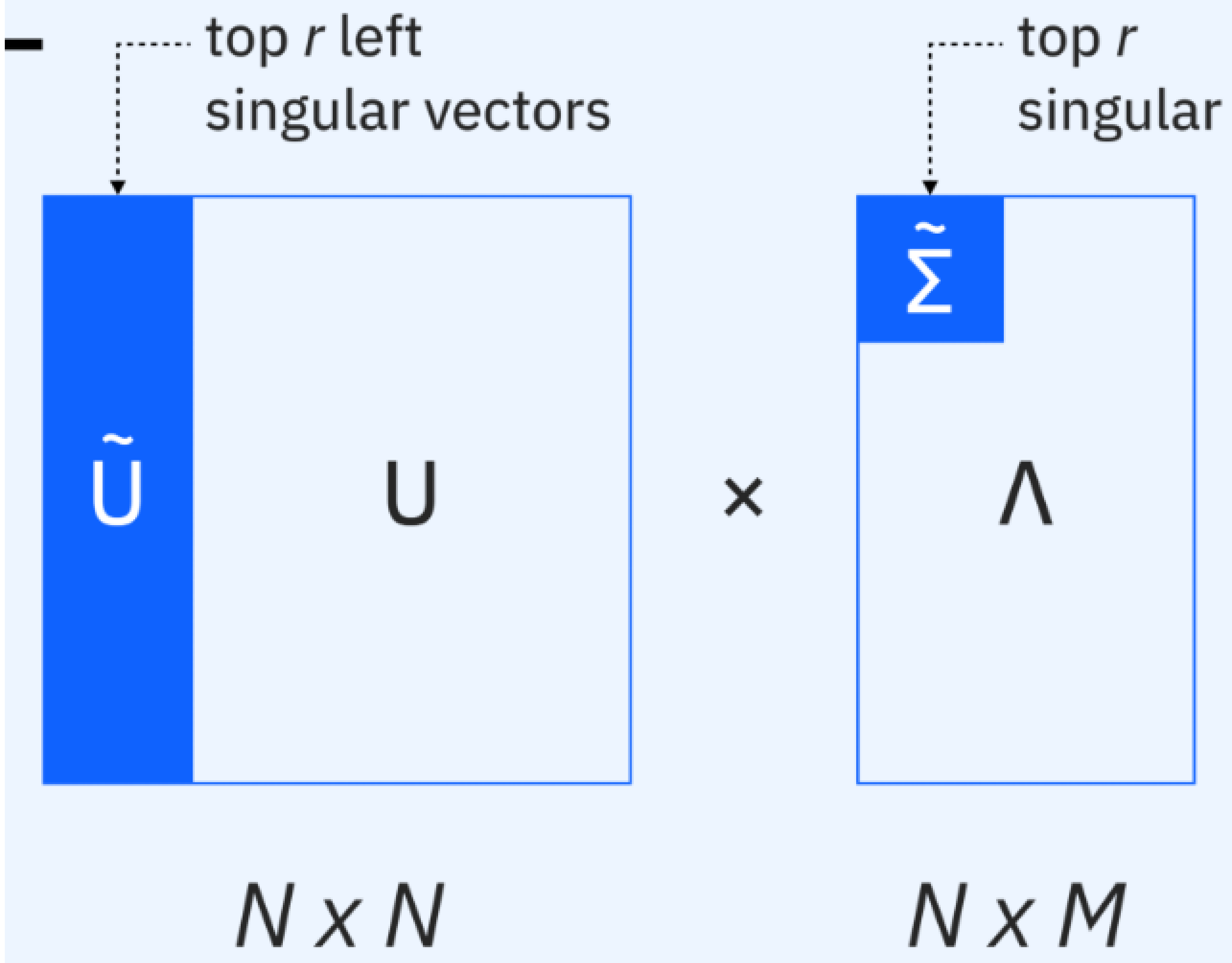
GloVe: Global Vectors

The Best of Both Worlds

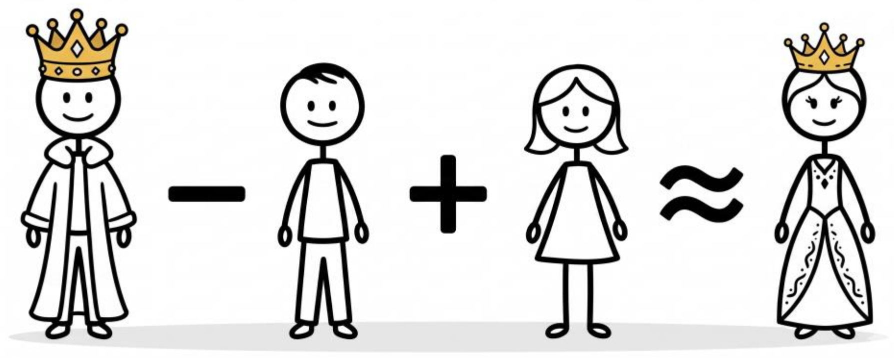
GloVe (Global Vectors for Word Representation) combines the global matrix factorization techniques of LSA with the local context window methods used in Word2Vec.

It assumes that meaningful relationships can be captured by the **ratio** of co-occurrence probabilities between words.

This allows it to efficiently leverage global statistics while maintaining the structural nuance of local word-to-word relationships.



The Algebra of Semantic Meaning



Vector Arithmetic

Dense representations like GloVe exhibit remarkable linear properties. Because semantic relationships are mapped to specific directions in space, we can perform arithmetic on meanings:

$$v(\text{King}) - v(\text{Man}) + v(\text{Woman}) \approx v(\text{Queen})$$

This proves that the dimensionality reduction process has successfully isolated **Gender** and **Royalty** as independent semantic dimensions.

WRAP-UP: PART 4

- ✅ The 3-step pipeline is the gold standard for semantic similarity.
- ✅ Statistics like TF-IDF and PPMI isolate signal from noise.
- ✅ Geometry (Cosine Similarity) allows magnitude-invariant comparison.
- ✅ SVD transforms sparse counts into meaningful dense embeddings.

MODERN MACHINE LEARNING ARCHITECTURE & THE FRONTIER

From Static Embeddings to Multimodal Alignment

PART 5

DENSE VS. SPARSE EMBEDDINGS

Sparse Representations

Traditional "Bag-of-Words" models create high-dimensional vectors (size of vocabulary). They are **sparse** (mostly zeros) and fail to capture semantic proximity.
Similarity is only recognized through exact keyword matches, leading to the "orthogonality trap."

Dense Semantic Neighborhoods

Modern NLP compresses meaning into **low-dimensional, real-valued vectors** (e.g., 300-768 dimensions).
Words with similar linguistic contexts occupy mathematically similar positions in the vector space, enabling semantic retrieval.

WORD2VEC (MIKOLOV ET AL.)

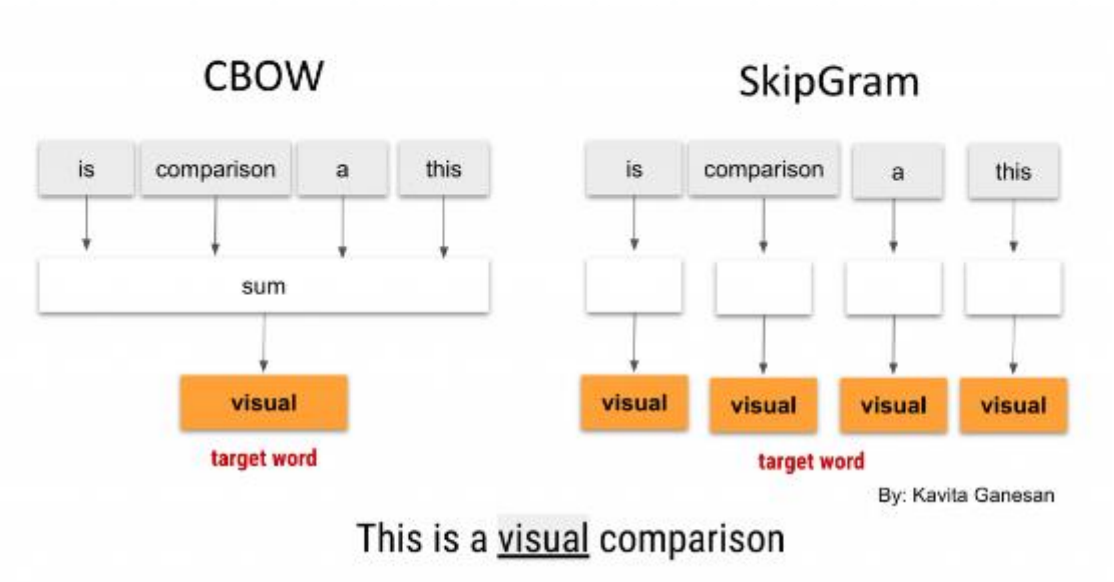
Static Distributed Meaning

Developed by Mikolov et al., Word2Vec uses shallow neural networks to learn representations.

- **CBOW:** Predicts a target word based on context words.
- **Skip-Gram:** Uses a single word to predict its surrounding context.

Ensures words appearing in similar contexts (e.g., "coffee" and "tea") cluster together.

$$P(w_c \mid w_t) = \frac{\exp(v_c^T v_t)}{\sum_{w \in V} \exp(v_w^T v_t)}$$



THE INFORMATION BOTTLENECK

The Polysemy Problem

Static models map one rigid vector to one word. This fails for **polysemy**—the "river bank" vs. "financial bank" paradox.

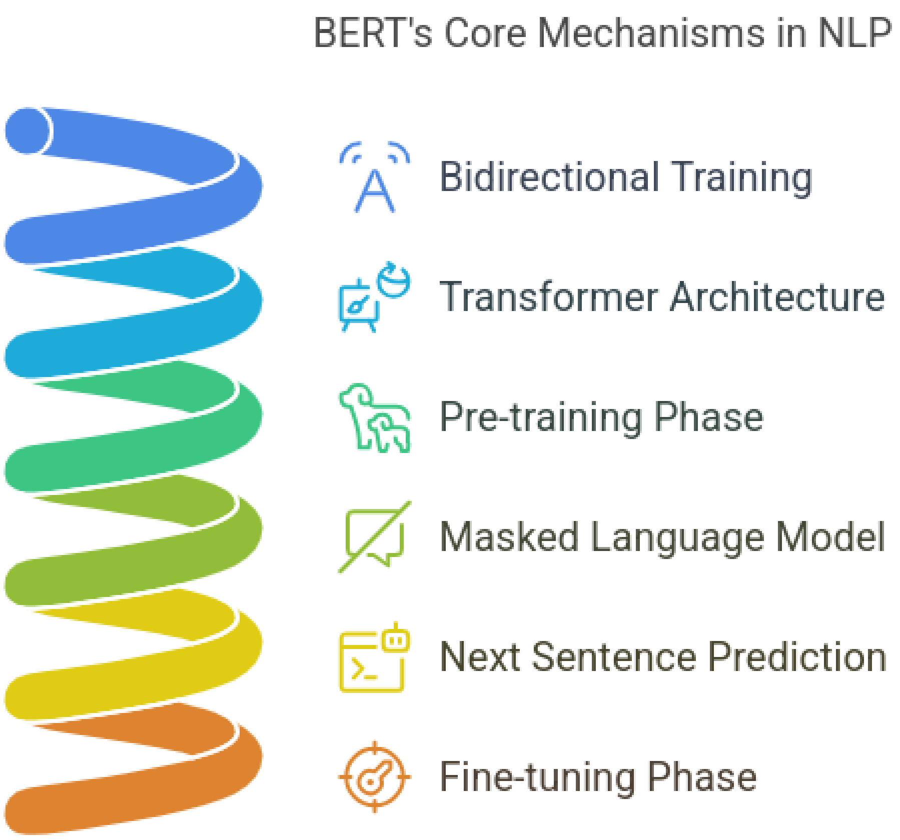
Nuanced meaning is lost because the model cannot adapt the vector to the specific sentence structure.

Dynamic Requirement

Human language is fluid. A single static lookup table is insufficient for the variability of meaning across diverse linguistic environments.

This gap necessitated the shift from *static* to *contextual* embeddings.

CONTEXTUAL EMBEDDINGS & BERT



Attention Mechanisms

Architectures like BERT (Bidirectional Encoder Representations fromTransformers) do not assign static vectors; representations change dynamically based on the surrounding sentence.

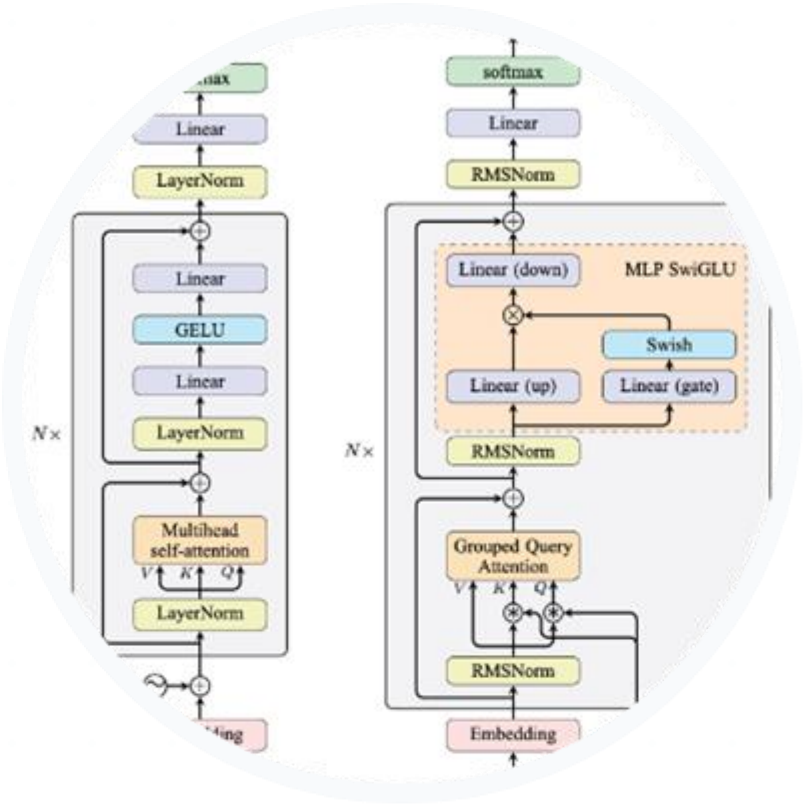
BERT reads text from both left and right simultaneously.

By leveraging the **Attention Mechanism**, it weighs the importance of every word in a sentence relative to every other word, resolving ambiguity dynamically.

BERT'S TRAINING MECHANISMS

- **Masked Language Model (MLM):** Randomly hiding 15% of tokens in a sequence, forcing the model to reconstruct them using bidirectional context.
- **Next Sentence Prediction (NSP):** Training the model to determine if two sentences naturally follow each other, capturing document-level coherence.
- **Fine-tuning Phase:** A small amount of task-specific labeled data is used to adapt the pre-trained knowledge to specific problems like sentiment or Q&A.

GPT & LANGUAGE MODELING



Next Token Prediction

LLMs like GPT are trained on the autoregressive process of predicting the subsequent word in a sequence.

By guessing billions of tokens, these "decoder-only" models inadvertently learn syntax, complex facts, and even logical reasoning chains.

Fluidity: Where BERT understands, GPT generates, building text word-by-word with human-like fluency.

COMPARING BERT AND GPT

Feature	BERT (Encoder-Only)	GPT (Decoder-Only)
Primary Objective	Masked reconstruction (NLU)	Next token prediction (NLG)
Context Flow	Bidirectional (Left & Right)	Unidirectional (Autoregressive)
Strength	Deep classification, Entailment	Creative generation, Dialogue
Use Case	Search ranking, Q&A systems	Coding, Essays, Chatbots

THE MULTIMODAL SHIFT

Beyond Text

Modern AI natively processes pixels, audio, and text simultaneously to bridge abstract symbols and the physical world.

Cross-Modal Alignment

Forcing different modality encoders (vision/text) into the exact same mathematical coordinate space.

Vision-Language (VLM)

Models like GPT-4o or LLaVA that can "see" an image and discuss it with the nuance of a human linguist.

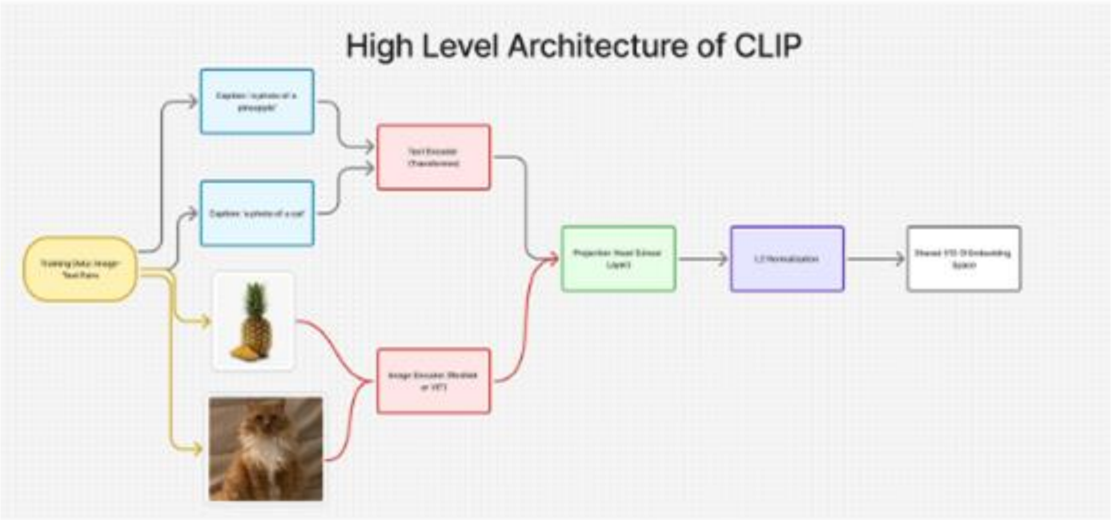
CLIP: SHARED LATENT SPACES

Contrastive Learning

CLIP (OpenAI) aligns images and text descriptions by maximizing their dot product in a shared space.

$$L = - \sum_{i=1}^{N} \log \frac{\exp(\mathrm{sim}(v_i, t_i)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(v_i, t_j)/\tau)}$$

This "shared latent space" allows the AI to recognize a "bear market" visually just as it does linguistically.



THE EVOLUTION SYNTHESIS

BoW Legacy	Embedding Era	Transformer Peak	Multimodal Future
Foundational but misses semantic nuance.	Word2Vec: Mapping words to fixed vectors.	BERT/GPT: Dynamic context and attention.	CLIP: Mapping fluid meaning across modalities.

Questions?

Thank you for your attention.

Valerio Modugno | May 2026